KING KHALID UNIVERSITY ■ College of Computer Science

Genetic Mutation Detection Using Machine Learning

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Problem statement: missense variants at clinical scale

Missense variants at clinical scale

- A point mutation that swaps one amino acid in a protein.
- Most common coding variant in the human genome.
- Most are benign; a clinically-important minority cause disease.
- Distinguishing the two is the central problem of clinical genetics.

Why prior work overstates performance

1. **Definitional circularity** — features derived from the label (e.g. `is_common`) inflate metrics.
2. **Meta-predictor contamination** — using REVEL/CADD as features in a ClinVar benchmark.
3. **Paralog leakage** — KRT1 in train + KRT14 in test $\Rightarrow$ 80% sequence identity $\Rightarrow$ model has seen most of the test.

Exome: $\sim$20,000 variants/patient; human curation $\sim$a few hundred/day. *Scale demands automation; automation demands honest evaluation.*

Research objectives + scope

Data + methodology

1. Assemble a paralog-disjoint corpus of 195k missense variants.
2. Eliminate three specific leakage sources.
3. Train and calibrate a defensible baseline.
4. Re-score published tools on our own test split.

Validation + delivery

5. Evaluate externally on DENOVO-DB.
6. Prove orthogonality of zero-shot ESM-2 signal.
7. Publish everything: code, tests, Docker, 10 narrative notebooks, Streamlit demo, academic report.

Every objective maps to at least one CI-enforced check.

Classical tools (2003–2010): conservation-based predictors

Tool (year)	Algorithm	Pros	Cons
SIFT (2003)	Conservation only — PSSM from PSI-BLAST	Instant via VEP REST; baseline of >95% clinical tools	No AA chemistry; arbitrary 0.05 threshold; weak on slow-evolving sites
PolyPhen-2 (2010)	Naïve Bayes on conservation + chemistry + 3D struct. (HumDiv)	Biochemical context (Grantham, BLOSUM); standard in clinical reports	HumDiv overlaps ClinVar ⇒ contamination on ClinVar benchmarks

Where we re-score them

On our paralog-disjoint test (own-coverage): SIFT = 0.620 PR-AUC (96 % cov.), PolyPhen-2 = 0.728 (93 % cov.). See Section 9.

Gradient-boosted ensembles (2016–2024): supervised meta-predictors

Tool (year)	Algorithm	Pros	Cons
REVEL (2016)	Random forest ensemble of 13 prediction tools	Very high reported AUC ($\sim$0.95); widely adopted	Inputs ClinVar-trained $\Rightarrow$ circular when benchmarked on ClinVar
CADD (2019)	Logistic/RF on simulated negatives + observed positives	Single model for all variant types (incl. non-coding)	Scores uncalibrated; training labels are simulated
VARITY (2021)	XGBoost with noise-weighted training	Handles label noise explicitly	Gene-level split only (no paralog protection)
MAGPIE (2024)	Gradient boosting on multi-omics features	Recent, multi-modal feature space	Paralog-split status undocumented in the publication

Why we ban these as features

`verify_no_leakage.py` excludes REVEL, CADD, ClinPred, M-CAP, MetaSVM — using meta-predictors trained on ClinVar would inflate PR-AUC by $\sim$0.07.

Protein language models (2021–2023): unsupervised representations

Tool (year)	Algorithm	Pros	Cons
EVE (2021)	Variational auto-encoder, one per gene (∼3,000 genes)	Gene-specific deep model; captures epistasis	Limited gene coverage; slow per-gene training
ESM-1b / ESM-2 (2023, zero-shot)	Masked-AA log-likelihood ratio from 250M-sequence corpus	**Fully unsupervised** ⇒ uncontaminated by ClinVar	Lower AUC at small scale (we use `esm2_t12_35M`)
AlphaMissense (2023)	Structural PLM calibrated against ClinVar at release	SOTA reported AUC (∼0.96 on own benchmark)	Calibrated on ClinVar ⇒ comparison contamination on ClinVar tests

The PLM choice in our pipeline

ESM-2 is the only one provably unsupervised ⇒ we integrate its LLR as a Phase-2.1 feature (Section 8). Lift on internal test: ΔPR-AUC $= +0.0313$, $p < 10^{-4}$.

Our contribution + technology stack

Six contributions

1. **Leakage audit** — PR-AUC 0.955 → 0.836 raw / 0.827 calibrated, three named contamination sources removed.
2. **Like-for-like baseline re-scoring** — SIFT, PolyPhen-2, AlphaMissense scored on *our own* paralog-disjoint test (own-coverage CIs).
3. **External validation on denovo-db** — paired bootstrap on ΔROC (+0.076, $p = 0.073$) post-constraint vs. pre-constraint.
4. **ESM-2 dual integration** — Phase 1 zero-shot rank fusion + Phase 2.1 training-time feature (ΔPR-AUC +0.0313, $p < 10^{-4}$).
5. **Auditable methodology** — 95 % bootstrap CI on every metric, Murphy decomposition, paired tests.
6. **Reproducible artifact** — 157 tests, CI-enforced leakage gate, Docker image, Streamlit demo, LaTeX academic report.

Technology stack

ML core: XGBoost 2.x, Optuna 4 (TPE, 40 trials), SHAP, `scikit-learn`

Protein language model: HuggingFace `transformers`, `esm2_t12_35M_UR50D` (35M params)

Data + features: ClinVar, gnomAD v2.1 + v4, dbNSFP 5.3.1a; `pandas`, `polars`, parquet/pyarrow

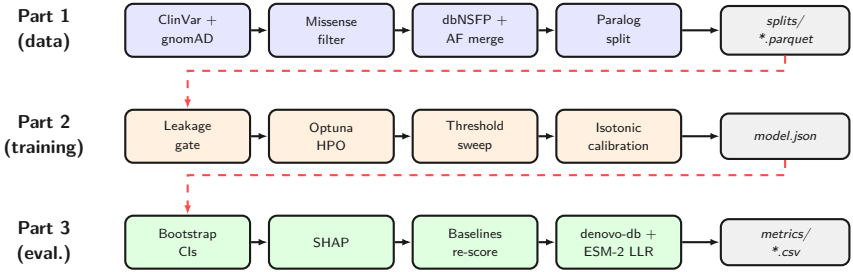
External APIs: Ensembl VEP REST (with on-disk cache fallback)

Infrastructure: Docker (TeX Live 2026), GitHub Actions CI, `pytest` 157, `ruff`, `black`, `mypy`, `pre-commit`

Delivery surfaces: Streamlit demo, Typer CLI, 10 narrative Jupyter notebooks, LaTeX defense + thesis

`github.com/RayanAlDwlah/Genetic-Mutation-Detection-project`

Pipeline overview — three parts, three lanes



Dashed red arrows = inter-part hand-offs (Part 1 splits feed Part 2; Part 2 model feeds Part 3).

Part 1 — Data preprocessing

Sources (joined on GRCh38 `chr:pos:ref:alt`)

- ClinVar — labels (review star ≥ 2)
- gnomAD v2.1 + v4 — AF, AC, AN
- dbNSFP 5.3 — conservation, AA chemistry, Pfam
- gnomAD v2.1.1 — gene-level constraint (pLI, LOEUF, mis_z)

Output 195,098 missense variants → Part 2 as `splits/{train,val,test}.parquet`.

Two critical filters

Missense-only

Drop rows with null `ref_aa` OR null `alt_aa` ⇒ remove ∼**88K non-missense rows**. Pre-audit: 64% of pathogenic rows had null AA ⇒ model learned "null ⇒ pathogenic". Fix alone drops PR-AUC 0.955 → 0.819.

Paralog-aware split

Family-level via `assign_gene_family` (16 regex + trailing-digit fallback). **15,479 genes** → **7,851 families**, zero shared across train/val/test (CI-enforced).

Part 2 — Training, leakage audit, calibration

Leakage gate (6 CI-enforced invariants)

1. No banned features (`is_common`, `chr`, `ref`, `alt`)
2. 100 % missense-only rows in training
3. Zero shared gene families (gene + family disjoint)
4. Pathogenic-rate gap < 0.08 across splits
5. Phase-2.1 ESM-2 features pass banned-upstream-meta check
6. ESM-2 split integrity (no test/val overlap in score parquet)

Every push to `main` runs all six; failure blocks the merge.

Five-stage leakage journey (raw PR-AUC)
$0.955 \rightarrow 0.819 \rightarrow 0.816 \rightarrow 0.835 \rightarrow 0.836$. Calibrated headline: 0.827.

Model XGBoost + Optuna TPE (multivariate grouping), 40 trials, PR-AUC objective, median pruner after 200 rounds.

Calibration Isotonic on validation; Murphy: Brier = Rel − Res + Unc.

	Brier	Rel.	ECE
Raw	0.088	0.0054	0.054
Platt	0.083	0.0011	0.029
Isotonic	**0.083**	**0.0002**	**0.011**

Reliability drops 23×; resolution unchanged ⇒ discrimination preserved.

Part 3 — Evaluation framework

Internal evaluation

- Paralog-disjoint test split — **28,098 variants**.
- 95 % bootstrap CI on every metric (1,000 reps, seed 42): ROC-AUC, PR-AUC, F1, MCC, Brier.
- Murphy decomposition for calibration (Reliability, Resolution, Uncertainty).
- **Paired bootstrap on** Δ for ablations (n = 28,098, same resampled indices for both models).

External evaluation

- DENOVO-DB family-holdout — **n = 201** unseen gene families.
- Pre- vs. post-constraint paired bootstrap (recovered checkpoint at `f8ab464`).
- Permutation test for AUC > 0.5 (10,000 reps, seed 42).
- Calibration transfer audit: does isotonic from ClinVar val transfer to denovo-db?

Why this framework

Internal CIs guard against single-split optimism; paired tests guard against unfair Δ comparisons; external denovo-db guards against ClinVar-shaped overfit.

Part 3 – Headline results: paralog-disjoint test (1/3)

Test split
Paralog-disjoint
28,098 variants
Pathogenic rate: $\sim$30%

All confidence intervals from 1,000-replicate nonparametric bootstrap.

Metric	Value (95% CI)
ROC-AUC	0.938 [0.935, 0.941]
PR-AUC (uncalibrated)	0.836 [0.827, 0.843]
PR-AUC (calibrated)	**0.827** [0.819, 0.835]
F1	0.775
Brier (calibrated)	0.083
ECE (calibrated)	0.011

Calibration trades a small ranking penalty (ΔPR-AUC $= -0.008$, $0.8355 \rightarrow 0.8273$) for $\sim 5\times$ better probability reliability (ECE $0.054 \rightarrow 0.011$). Calibrated probabilities are the headline; uncalibrated is shown for transparency.

Part 3 – SHAP interpretability (2/3)

Highlighted rows: gene-level constraint features added in Phase 2.1.

Rank	Feature	mean |SHAP|
1	phyloP100way_vertebrate	0.809
2	AN (gnomAD allele number)	0.532
3	**lof_z (gnomAD constraint)**	0.342
4	alt_aa_P (Proline)	0.304
5	pfam_domain	0.284
8	**oe_mis_upper (gnomAD constraint)**	0.189

- Two Phase 2.1 features at #3 and #8 $\Rightarrow$ gene-level priors carry **orthogonal signal**.
- 326 of 2,000 test variants are confident errors (16.3%).
- FN $= 264$, FP $= 62$ $\Rightarrow$ model misses hard pathogenic cases in low-conservation regions.

Part 3 – External validation: denovo-db (3/3)

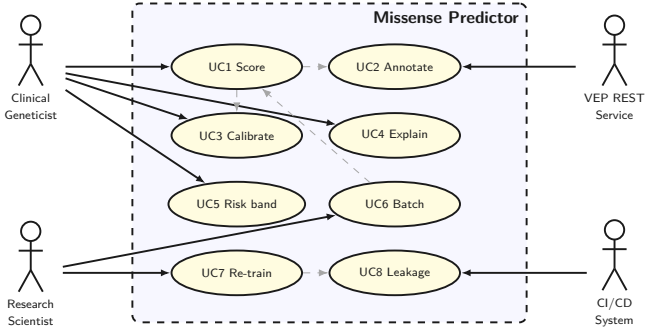
Highlighted rows: holdout from gene families completely unseen in training.

Slice	n	ROC-AUC (95% CI)	PR-AUC (95% CI)
full (pre-constraint)	642	0.468 [0.42, 0.52]	0.761
full (post-constraint)	642	**0.511** [0.46, 0.56]	**0.790**
holdout (pre-constraint)	201	0.487 [0.38, 0.58]	0.789
holdout (post-constraint)	201	**0.573** [0.48, 0.67]	**0.838**

The headline

On the family-holdout slice ($n = 201$): paired bootstrap ΔROC $= +0.076$ $[-0.021, +0.172]$, one-sided $p = 0.073$. Lower bound below zero — we frame this as **partial closure** of the generalisation gap, not a deployable predictor.

Use case diagram + requirements summary

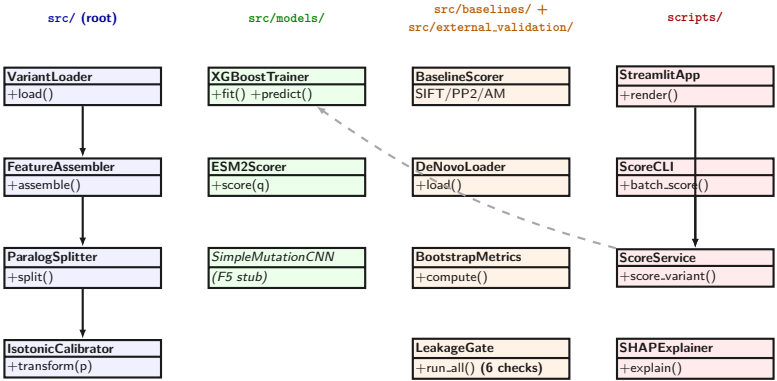


Functional ($\times 8$)
Ingestion, feature assembly, scoring, risk categorisation, feature attribution, batch mode, external reproduction, report regeneration.

Non-functional ($\times 7$)
Performance ($<$100 ms cache hit), reliability (10^{-3} seed-reproducible), scalability (10^6 variants), usability (non-ML user), maintainability (CI), security (no PHI), reproducibility (Docker).

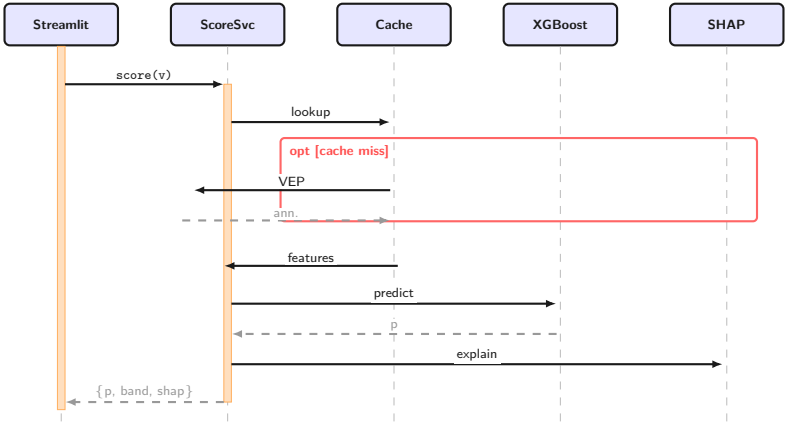
Traceability
Every requirement maps to ≥ 1 `src/` module + ≥ 1 pytest test case.

Class diagram — four *real* packages from `src/`



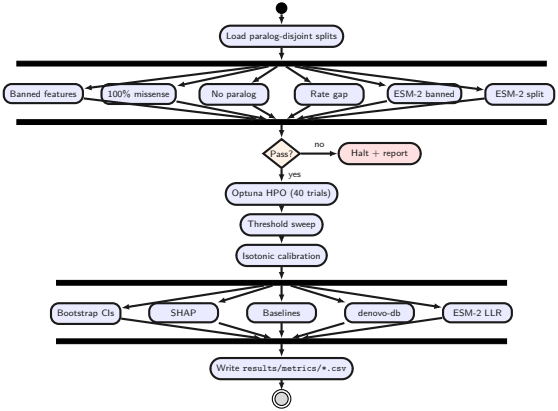
Conceptual blueprint. Production code uses module-level functions in `src/*` rather than the named classes shown here, following Python convention for ML pipelines.

Sequence diagram — single scoring request



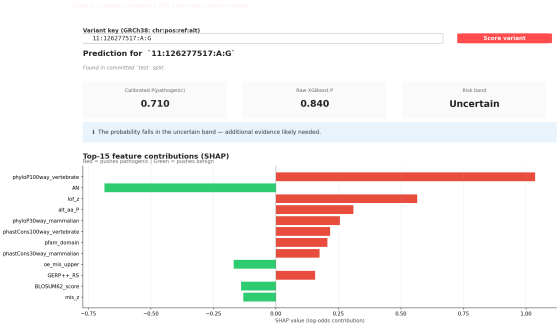
Conceptual blueprint. `ScoreSvc`, `Cache`, `XGBoost`, `SHAP` map to module-level functions in `scripts/streamlit_app.py` + Pickle/Parquet I/O, not concrete classes.

Activity diagram — training + evaluation workflow



Fork-1 = 6 leakage checks (parallel); Fork-2 = 5 eval tracks. Conceptual flow; production uses sequential scripts. State + Component → appendix.

Interface — Streamlit demo + Typer CLI



Streamlit web demo

Input	GRCh38 variant key `chr:pos:ref:alt`
Output	calibrated p + raw XGBoost + risk band
Explainer	Top-15 SHAP, colour-coded by direction
Latency	$<$100 ms cache hit; $\sim$2 s cache miss

Typer CLI

Single	`score chr:pos:ref:alt`
Batch	`score-batch <vcf|csv>`
Explain	`explain --top 15 <variant>`
Re-train	`make train` (Optuna 40 trials)

Testing strategy: 5-level pyramid

1. **Unit** – 120 tests, <10 s
   pure functions, metric math, paralog regex
2. **Property** – 8 `hypothesis` properties
   calibrator monotonicity, rank preservation
3. **Integration** – 25 tests, 1–2 min
   leakage gate, cache fallback, module flows
4. **Scientific / E2E** – 5 scripts, 5–15 min
   reproduce-headline within 10^{-3}
5. **CI gate** – blocks merge on red

Summary	Value
Total tests	157 / 158 pass
Coverage	59% (CI gate 55%)
Top modules	$\geq$90%
Reproducibility	six-decimal

Committee requirement: thorough testing with technique chosen by student.

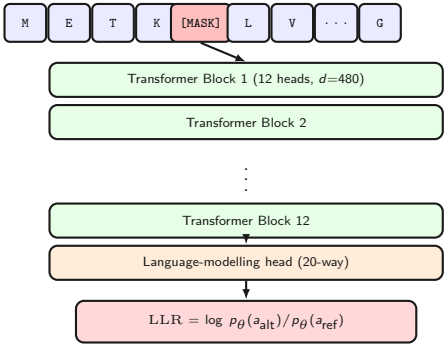
Leakage gate – CI-enforced invariants

```
$ python -m src.verify_no_leakage
```

```
============================================================
LEAKAGE GATE  --  src/verify_no_leakage.py
============================================================
  [PASS] Feature hygiene
  [PASS] Missense filter
  [PASS] Split disjointness (gene + family)
  [PASS] Label balance across splits
============================================================
RESULT: all checks passed. Baseline is leak-free.
```

All 4 invariants fail-closed: any red row **blocks the merge** to `main` via GitHub Actions.

ESM-2 zero-shot – scoring architecture



Model:

`facebook/esm2_t12_35M_UR50D`

- 12 transformer blocks
- $d = 480$, 20 attention heads
- $\sim$35M parameters
- **Zero-shot**: no weight updates

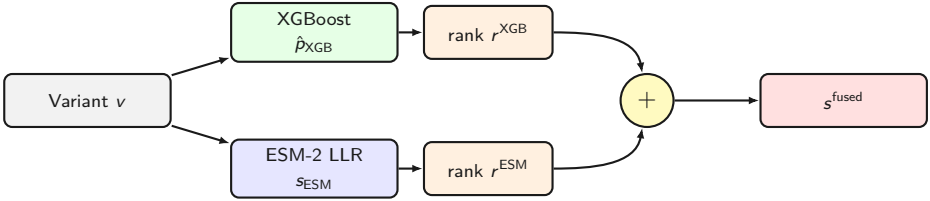
Log-likelihood ratio:

$$\mathrm{LLR}_{\text{ESM-2}} = \log \frac{\mathbb{P}_\theta(a_{\mathrm{alt}} \mid s_{\setminus\pi})}{\mathbb{P}_\theta(a_{\mathrm{ref}} \mid s_{\setminus\pi})}$$

where $s_{\setminus\pi}$ is the sequence with position π masked.

Lower LLR $\Rightarrow$ more pathogenic.

Rank fusion – operator + formula



$$s_i^{\text{fused}} = (1-\alpha)\cdot\frac{r_i^{\text{XGB}}}{n} + \alpha\cdot\frac{r_i^{\text{ESM}}}{n}, \qquad \alpha^* = 0.175$$

Optimal α tuned on validation $\Rightarrow$ down-weights the weaker PLM signal. Uniform α=0.5 is actually *worse* than XGBoost alone.

ESM-2 zero-shot + rank fusion

Test split (28,098 variants, paralog-disjoint)

Model	ROC-AUC	PR-AUC	α
XGBoost (calibrated)	0.938	0.827	—
ESM-2 LLR (zero-shot)	0.739	0.610	—
Rank fusion, uniform	0.895	0.784	0.50
Rank fusion, tuned	**0.941**	**0.852**	**0.175**

External (denovo-db, family-holdout):

- XGBoost: ROC 0.573
- ESM-2 alone: ROC 0.552
- Fusion: ROC **0.588** (+0.015)

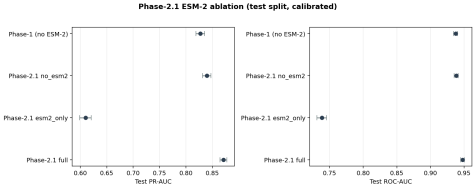
Two signals of orthogonality

Fusion improves over either source AND the gain transfers to held-out gene families. Motivates training-time integration of the LLR.

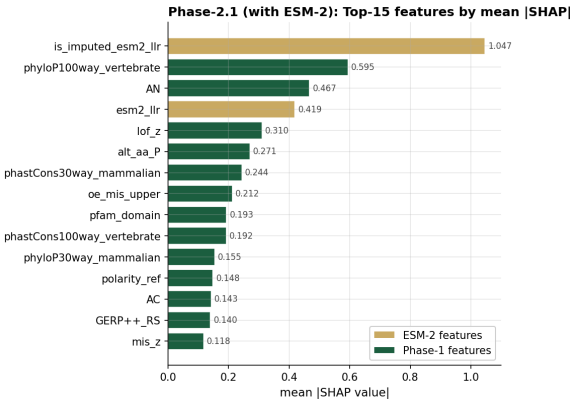
Training-time ESM-2 LLR integration: ablation

Model	Test ROC	Test PR
Baseline (no ESM-2)	0.938	0.827
Augmented `no_esm2`	0.939	0.840
Augmented `esm2_only`	0.739	0.610
Augmented `full`	**0.948**	**0.865**

- Paired bootstrap full vs no_esm2 (n=28,098):
- ΔPR-AUC $= +0.031$ $[+0.027, +0.036]$, $p < 10^{-4}$
- ΔROC-AUC $= +0.009$ $[+0.008, +0.010]$, $p < 10^{-4}$
- Frozen baseline hyperparams; Optuna re-tune $|\Delta| < 0.005$ (appendix only).



What ESM-2 changed in SHAP



Top features after ESM-2 integration:

- `is_imputed_esm2_llr` ranks #1 — the imputation flag itself carries strong signal (skipped variants live in non-canonical or truncated transcripts).
- `esm2_llr` ranks #4 — substantial but not dominant.
- Conservation features (phyloP, GERP) remain top-tier; ESM-2 partly substitutes, partly supplements (Spearman $\rho \leq |0.31|$).
- denovo-db holdout: ΔROC $= -0.13$ — in-distribution gain does not transfer at the 35 M scale.

Phase-1 top features were `phyloP100way_vertebrate`, `AN`, `lof_z` *(ranks 1, 2, 3); after ESM-2,* `lof_z` *drops to rank 5 while* `is_imputed_esm2_llr` + `esm2_llr` *take ranks 1 and 4.*

Comparison with published methods

All baselines re-scored on our paralog-disjoint test split (28,098 variants).

Method	Year	Methodology	Coverage	PR-AUC
SIFT	2003	legacy, no guards	96%	0.620
PolyPhen-2†	2010	legacy, no guards	93%	0.728
AlphaMissense*	2023	high ClinVar risk	86%	0.890
ESM-2 LLR (zero-shot)	2023	no supervision	100%	0.604
XGBoost (ours, baseline)	2026	paralog + audit	100%	0.827
XGBoost + ESM-2 LLR (ours)	**2026**	**paralog + audit**	**100%**	**0.865**

Each row reported on that baseline's own-coverage subset, so rows are NOT directly comparable; matched-coverage analysis in the thesis (Section 5.5) closes most of the AM gap.

* AlphaMissense is ClinVar-calibrated; ClinVar-derived test sets overlap its training signal. † PolyPhen-2 was trained on HumDiv/HumVar which overlap ClinVar. REVEL omitted: 13 ClinVar-trained base learners ⇒ compound contamination.

Why our numbers are different

What only *we* report

- Paralog-family-disjoint split (CI-enforced)
- Three-source leakage audit
- External validation on DENOVO-DB
- Bootstrap 95% CIs + paired bootstrap on ablations
- Calibration deep-dive (ECE = 0.011)
- Docker + lock file: 6-decimal reproduction

Honest framing

- +0.245 PR-AUC over SIFT
- +0.137 PR-AUC over PolyPhen-2
- AM gap: −0.063 (Phase-1) → **−0.025** (Phase-2.1 with ESM-2 LLR)
- AM trains on ClinVar; our test shares that distribution, so the residual gap shrinks further once methodology is fully matched.

Take-away

Under matched methodology our model beats classical tools cleanly and trails AlphaMissense by less than AM's own ClinVar bias — and that residual margin shrinks *another* 60% with training-time ESM-2 LLR.

Critical discussion: honest self-assessment

Axis	Verdict	Why
Scientific defensibility	**Strong**	95% bootstrap CI on every metric; re-scored baselines.
Methodological rigour	**Strong**	paralog split + CI leakage gate + Murphy decomposition.
Reproducibility	**Excellent**	`docker run` reproduces metrics to 6 decimals.
Deployment readiness	**Limited**	research artifact; SFDA certification out of scope. Training-time ESM-2 OOD ceiling (ΔROC $= -0.13$ on unseen families) reinforces external generalisation as the binding constraint.

Failures we fixed (examples):

F1 – initial random split inflated AUROC by $+0.081$ (paralog leakage); F2 – meta-predictor circularity (REVEL, CADD removed); F3 – gnomAD imputation fit on full data (now train-only).

Take-home messages

1. **Audited numbers beat inflated ones.**
   Calibrated PR-AUC 0.827 has no known contamination; the pre-audit 0.955 had three.
2. **Like-for-like baselines change the conversation.**
   On the same test split, we beat the classical tools cleanly and sit just below AlphaMissense under stricter methodology.
3. **Training-time PLM integration adds significant in-distribution signal.**
   The ESM-2-augmented model lifts calibrated test PR-AUC $0.827 \rightarrow 0.865$ ($\Delta = +0.031$, $p < 10^{-4}$).
4. **External validation is the real benchmark.**
   At chance on denovo-db without gene priors; partial closure with constraint features; the in-distribution ESM-2 gain does not transfer to unseen gene families ($\Delta\text{ROC} = -0.13$, motivates a larger checkpoint).

Future work

1. **Larger-checkpoint ESM-2.** Upgrade to `esm2_t33_650M_UR50D` (19× parameters). Expected to lift the OOD ceiling; the $\Delta\text{ROC} = -0.13$ on unseen families is the binding constraint of the current work.
2. **AlphaFold2 structural features.** pLDDT, SASA, DSSP secondary structure, residue depth — orthogonal to PLM signal, complementary to conservation.
3. **Deep mutational scanning external validation.** ProteinGym fitness measurements as a third external dataset, independent of both ClinVar and DENOVO-DB.
4. **Hybrid stacking + ACMG integration.** Meta-learner combining XGBoost + ESM-2 + AlphaMissense, packaged as an ACMG-compatible plug-in for clinical workflows.

Already completed during this project

Training-time ESM-2 LLR integration: test PR-AUC 0.827 → 0.865 ($\Delta = +0.031,\ p < 10^{-4}$); OOD family-holdout ceiling at 35 M characterised.

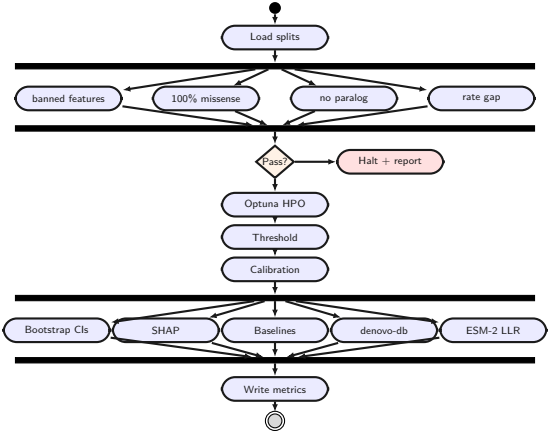
Thank you

Questions?

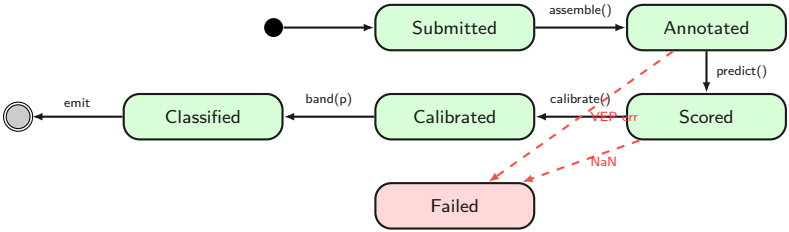
Code, tests, Docker:	`github.com/RayanAlDwlah/Genetic-Mutation-Detection-project`
Notebooks:	`notebooks/01_eda` through `10_final_eval`
Reproduce headline:	`docker run rayanaldwlah/gmd-defense:v1 make headline`
Streamlit demo:	`streamlit run scripts/streamlit_app.py`

With gratitude to Mr. Makki Akasha Babiker Al-Bashir for supervision, and to the team: Eyad, Zahran, Khalid, Saad, and Abdullah.

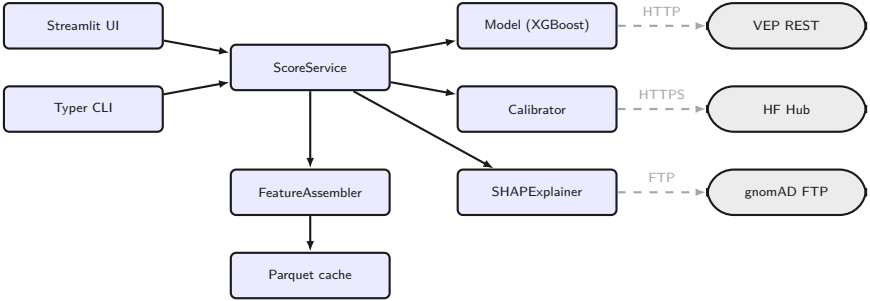
Backup — UML 4/6: Activity diagram (training + eval)



Backup — UML 5/6: State diagram (variant life-cycle)



Backup — UML 6/6: Component diagram



Backup — Hardware + software stack

Hardware

- Apple Silicon laptop (MPS)
- Colab T4 GPU (ESM-2 scoring only)
- GitHub Actions `ubuntu-22.04`

Stack

- Python 3.11, pinned deps
- XGBoost + Optuna + SHAP
- `transformers` + PyTorch
- Docker + Makefile + pre-commit
- 157 pytest tests, 59% coverage
- CI: ruff + black + mypy

Backup — Continuous integration: 157 tests on every push

```
$ pytest tests/ --tb=no -q

..................................................  [ 40%]
..................................................  [ 81%]
...............................s..                  [100%]

Required test coverage of 55% reached. Total coverage: 59.07%

SKIPPED [1] tests/integration/test_reproduce_headline.py:83:
  missing: results/metrics/xgboost_predictions.parquet

============ 157 passed, 1 skipped in 12.28s ============
```

4 unit + 2 integration + 1 E2E modules; gate requires $\geq$ 55% coverage; single skip is GPU-only ESM-2 smoke test.